

JABESS OMANE

AI Research Engineer | Responsible AI, Human-AI Interaction and Policy

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[Portfolio](#) | [LinkedIn](#)

RESEARCH PROFILE

AI research engineer working on explainable machine learning and counterfactual recourse. Research interests connect responsible AI, human-computer interaction, tech and AI policy, and the economic effects of technology, with attention to African contexts. Training in policy-facing ML and agent-based modelling supports this interdisciplinary approach.

EDUCATION

BTech Mechatronics Engineering | First Class Honours

Koforidua Technical University, Ghana | 2021 - 2024

Selected coursework: Calculus, Computer Programming, Control Systems, Embedded Systems, Digital Signal and Image Processing, Robotics, Research Methods.

SELECTED RESEARCH AND PROJECTS

FairScore | Transparent Algorithmic Risk Assessment | Ongoing

- Built a risk-scoring system using gradient-boosted decision trees and ensemble methods for high-stakes behavioral data.
- Applied SHAP and LIME to investigate global and local model behavior and support exploratory model validation.
- Implemented counterfactual recourse with DiCE-ML to communicate how alternative inputs could change model predictions.
- Investigated confounding and proxy discrimination using causal analysis in automated decision pipelines.

DecisionLens | Counterfactual Recourse for Graduate Admissions | Research prototype

- Developed an admissions prototype combining a random forest classifier with DiCE counterfactual generation.
- Compared baseline and feature-cost-weighted recourse, with downstream adjustments under manually specified causal scenarios.
- Built applicant-facing explanations and explored recourse validity across academic-background groups; examined limitations of proxy outcomes and assumed action costs.

HexAI | Ghanaian Sign Language Translation | Ongoing

- Contributed to a bidirectional sign-language AI project, coordinating between deaf and hard-of-hearing communities and software, AI, and design teams.
- Prepared a data card and contributed technical work to support accessible interaction and dataset documentation.

RESEARCH AND PROFESSIONAL EXPERIENCE

AI Research Engineer | Mingo Blox | Accra, Ghana | 2025 - Present

- Lead applied AI research and prototyping across education, telecommunications, and industry use cases, with attention to explainability, fairness, safety, and usability.
- Communicate technical findings to engineers, founders, partners, and non-technical stakeholders.
- Represented Mingo Blox in the 2025 UNICEF Startup Lab cohort, supported by KOICA and UNICEF Ghana.

TALKS AND RESEARCH COMMUNICATION

Speaker | PyTorch Community East Africa | Online | 29 August 2026

AI Safety: Building Safer and More Responsible AI Systems

- Presented practical AI safety topics including interpretability, bias and fairness, hallucinations, alignment, misuse, security, and open-weight model considerations.
- Discussed evaluation, community participation, and responsible deployment in African contexts.

[Organizer recap](#) | [Event listing](#)

GOVERNANCE AND COMMUNITY ENGAGEMENT

Ghana Internet Governance Fellow | 2026

- Selected for the Ghana Internet Governance Fellowship; participate in discussions on digital policy, internet governance, and responsible technology development.

Ambassador | Thinking About Thinking | 2026

- Engage in interdisciplinary discussions connecting AI, neuroscience, cognitive science, and human-centered system design.

SELECTED ADDITIONAL TRAINING

Online study and professional development

Introduction to Agent-Based Modelling | UNU-MERIT / Maastricht University | 2026

Python, R, and NetLogo; socioeconomic simulation, model validation, and sensitivity analysis.

ARENA learning materials | 2026

Fundamentals, Transformer Interpretability, and LLM Evaluations.

ML Safety | ML Safety.org | 2026

Robustness, monitoring, systemic risk, and safe deployment.

AI Research Foundations | Google DeepMind learning pathway | 2025

Neural networks, transformers, language modeling, fine-tuning, and model evaluation.

Mathematics of Machine Learning Summer School | The Alan Turing Institute | 2025

Supervised learning, stochastic optimization, generalization, and statistical learning.

Assessing and Mitigating Bias in AI | The Alan Turing Institute | 2025

Fairness assessment and mitigation, including topics beyond binary classification.

Machine Learning for Public Policy | UNU-MERIT / Maastricht University | 2025

Machine learning methods, bias, and policy-focused evaluation.

TECHNICAL SKILLS

Programming and data: Python, R, C, C++, SQL, Pandas, NumPy.

Machine learning: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM; predictive modeling, ensemble methods, hyperparameter tuning, and model evaluation.

Interpretability and fairness: SHAP, LIME, DiCE-ML, AIF360, Fairlearn; counterfactual recourse, fairness evaluation, and LLM evaluation.

Language models: Hugging Face Transformers, fine-tuning, LoRA, PEFT, quantization.

Causal analysis and experiments: DoWhy, EconML, CausalML, MLflow, Weights & Biases.